

Learning from N -Tuple Data with M Positive Instances: Unbiased Risk Estimation and Theoretical Guarantees

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Abstract—Weakly supervised learning often operates with coarse aggregate signals rather than instance labels. We study a setting where each training example is an n -tuple containing exactly m positives, while only the *count* m per tuple is observed. This NTMP (N -tuple with M positives) supervision arises in, e.g., image classification with region proposals and multi-instance measurements. We show that tuple counts admit a *trainable unbiased risk estimator* (URE) by linking the tuple-generation process to latent instance marginals. Starting from fixed (n, m) , we derive a closed-form URE and extend it to variable tuple sizes, variable counts, and their combination. Identification holds whenever the effective mixing rate is separated from the class prior π (i.e., $\bar{\alpha} \neq \pi$). We establish generalization bounds via Rademacher complexity and prove statistical consistency with standard rates under mild regularity assumptions. To improve finite-sample stability, we introduce simple ReLU corrections to the URE that preserve asymptotic correctness. Across benchmarks converted to NTMP tasks, the approach consistently outperforms representative weak-supervision baselines and yields favorable precision-recall and F_1 trade-offs. It remains robust under class-prior imbalance and across diverse tuple configurations, demonstrating that count-only supervision can be exploited effectively through a theoretically grounded and practically stable objective.

Index Terms—weakly supervised learning, unbiased risk estimation, generalization bounds

I. INTRODUCTION

The rapid progress of modern AI has intensified the demand for large, labeled corpora [1]. Fully supervised pipelines [2]–[4] hinge on exhaustive instance-level annotation, which is costly or infeasible in sensitive domains such as healthcare and scientific experiments [5], [6]. This has motivated a rich literature on weakly supervised learning (WSL) from incomplete, inexact, or noisy supervision [7]–[10], including positive-unlabeled (PU) learning [11]–[14], partial-label learning [15], [16], complementary-label learning [17]–[20], positive-confidence variants [21], [22], UU learning [23], [24], and tuple-/relation-based supervision using similarities, confidences, and comparisons [25]–[27].

A recurring practical theme is that *aggregate* labels are easier to collect than instance labels. Labels from Label Proportions (LLP) replace per-instance annotation with bag-level class proportions and can be learned via distributional constraints [28], [29]. Multi-instance learning (MIL) similarly

exploits bag-level signals [30], [31]. However, a known limitation of *pure* LLP is *non-identifiability when all bags share the same class proportion*: if every bag has the identical positive rate, the constraints collapse to matching a single global mean, leaving many degenerate solutions and little instance-level supervision. By contrast, many applications provide an *exact positive count* per tuple (e.g., an image contains exactly m positive regions among n proposals). This signal retains the collection/privacy advantages of LLP while differing structurally: the tuple size n and the *exact* count m are known, but positions are unobserved.

In light of these observations, we focus on the exact-count variant of aggregate supervision. Rather than treating it as yet another supervision type parallel to LLP/MIL, we ask: *Can tuple-level counts be turned into a principled, trainable objective with finite-sample control, even when across-tuple variation is minimal?* This perspective motivates the NTMP formulation below.

We study learning from n -tuples that contain exactly m positives (NTMP), and denote the within-tuple positive proportion by $\alpha := m/n$. Rather than focusing on supervision type alone—which conceptually overlaps with LLP/MIL—we develop a *learning principle* that turns tuple-count information into a tractable *unbiased risk estimator* (URE) with finite-sample control. Crucially, *even if all tuples share the same proportion α* (i.e., no across-tuple variation), our formulation remains identifiable and trainable *provided $\alpha \neq \pi$* , where π is the population class prior. In practice this mild separation can be ensured by design (e.g., selecting n or constructing tuples so that α does not coincide with π). We quantify sensitivity to this gap and show that stability scales with $|\pi - \alpha|^{-1}$; simple clamps keep the empirical objective well-behaved as $\alpha \rightarrow \pi$ (see Section IV). To operationalize learning, we pair NTMP tuples with an unlabeled reference set whose class prior is treated as known (an *unlabeled- π pool*). The prior can be obtained from domain statistics or standard prior-estimation/calibration on a small validation batch; we also assess misspecification in Section IV. This pairing yields a closed-form linear system that eliminates the unknown class-conditionals and results in a URE amenable to standard ERM.

Our contributions:

- **Unbiased risk from tuple counts + an unlabeled- π pool.** We formalize the NTMP data generation, show that flattening tuples yields a mixture with positive rate $\alpha = m/n$, and combine this with an unlabeled set of known prior to derive a closed-form linear system that removes

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class-conditionals from the population risk, resulting in a practical URE (no instance labels required).

- **Variance-optimal intra-tuple weighting.** Within the URE, we analyze per-tuple weightings and show that uniform instance weighting inside each tuple minimizes estimator variance under standard i.i.d. assumptions, providing a principled default for stochastic optimization.
- **Generalization guarantees and consistency.** Using Rademacher complexity, we establish generalization bounds and prove consistency (up to standard constants) under NTMP supervision.
- **Practical stability corrections.** As in other weak-supervision settings, the unbiased objective can be high-variance. We incorporate simple non-negative corrections (ReLU/ABS clamps) that mitigate overfitting while preserving asymptotic correctness.

LLP and MIL demonstrate the value of aggregate supervision; our formulation targets the *exact-count* variant on fixed or varying (n, m) and provides a direct, closed-form URE that requires only counts and an unlabeled- π reference—no pairwise identities, orders, or confidences [25]–[27], [32]. Importantly, identifiability hinges only on $\alpha \neq \pi$ (rather than on inter-bag proportion variation), thus avoiding a core failure mode of pure LLP where equal bag proportions lead to degeneracy.

On image benchmarks (MNIST, FASHIONMNIST, SVHN, CIFAR-10) converted to NTMP tasks, our approach improves over UU and clustering baselines, and stability corrections further curb overfitting. We also examine robustness to class-prior shifts and tuple configurations, reflecting the practical benefits of aggregate labels while offering a tailored objective for exact-count supervision (see Section IV).

Roadmap: Section II reviews preliminaries and related work. Section III introduces the NTMP model, derives the URE, presents generalization results, and gives the training algorithm with stability corrections. Section IV reports experiments and robustness analyses. Section V concludes and discusses extensions (e.g., variable (n, m) and tuple-aware architectures).

II. PRELIMINARIES

This section fixes notation, recalls the supervised learning setup, and positions our problem within related weakly supervised paradigms.

A. Supervised Learning

We consider binary classification with feature space $\mathcal{X} \subset \mathbb{R}^d$ and label space $\mathcal{Y} = \{+1, -1\}$. Let $(\mathbf{x}, y) \sim p(\mathbf{x}, y)$ denote a generic example from an unknown distribution. Write the positive-class prior as

$$\pi := \mathbb{P}(y = +1) \in (0, 1), \quad 1 - \pi = \mathbb{P}(y = -1),$$

and let the class-conditionals be $p_+(\mathbf{x}) := p(\mathbf{x} \mid y = +1)$ and $p_-(\mathbf{x}) := p(\mathbf{x} \mid y = -1)$. The marginal over features decomposes as

$$p(\mathbf{x}) = \pi p_+(\mathbf{x}) + (1 - \pi) p_-(\mathbf{x}).$$

A classifier is a scoring function $g : \mathcal{X} \rightarrow \mathbb{R}$ (the sign of $g(\mathbf{x})$ predicts the label). Given a loss $\ell : \mathbb{R} \times \{+1, -1\} \rightarrow \mathbb{R}_+$ (e.g., 0–1, logistic, hinge), the population risk is

$$\begin{aligned} \mathcal{R}(g) &= \mathbb{E}_{(\mathbf{x}, y) \sim p} [\ell(g(\mathbf{x}), y)] \\ &= \pi \mathbb{E}_{p_+} [\ell(g(\mathbf{x}), +1)] + (1 - \pi) \mathbb{E}_{p_-} [\ell(g(\mathbf{x}), -1)]. \end{aligned} \quad (\text{II.1})$$

In fully supervised learning, one minimizes an empirical approximation of (II.1) using labeled samples. In our weakly supervised setting, pointwise labels are unavailable; we will express $\mathcal{R}(g)$ using tuple-level count signals together with an unlabeled reference pool (Section III).

B. Related Work

UU learning.: UU learning trains a binary classifier from two unlabeled datasets whose class priors differ and are known (or estimable) [23], [24]. Let $\mathcal{U}_1, \mathcal{U}_2$ be unlabeled sets drawn from mixtures with priors $\pi_1 \neq \pi_2$; then (p_+, p_-) can be recovered via a 2×2 linear system, yielding an unbiased risk estimator akin to PU learning. Our NTMP setting is distinct but connected: if we *flatten* all tuples into individual instances, we obtain an unlabeled pool whose effective positive rate equals the tuple ratio $\alpha := m/n$. By coupling this pool with an additional unlabeled dataset whose prior π is known, we obtain an analogous linear system eliminating (p_+, p_-) and leading to a tractable unbiased risk (Section III). This perspective places NTMP within the broader family of risk-reconstruction methods that exploit mixtures with differing priors [23], [24].

Pairwise similarity/dissimilarity supervision.: Another related line learns from pairwise same-label (SU) or different-label (DU) constraints. Here the weak labels reside in

$$\mathcal{Y}_S = \{(+1, +1), (-1, -1)\}, \quad \mathcal{Y}_D = \{(+1, -1), (-1, +1)\},$$

and an unlabeled distribution is typically used to identify class marginals. Our NTMP formulation generalizes the pairwise cases to n -tuples with *known positive counts* m : when $(n, m) = (2, 1)$ it recovers dissimilar pairs; when $(n, m) \in \{(2, 0), (2, 2)\}$ it recovers similar pairs. Unlike methods that require pair identities, orders, or confidence scores, NTMP uses only count-level constraints and still admits an unbiased-risk objective once coupled with an unlabeled reference.

Learning from Label Proportions (LLP).: LLP assumes bags of unlabeled instances with observed *class proportions* per bag, aiming at instance-level prediction. Early work established proportion-based formulations and consistency under identifiable mixture assumptions [33]; max-margin variants (often termed proportion-SVM) optimize latent labels subject to proportion constraints [34]; other approaches cast LLP as constrained clustering/partitioning [35]; recent analyses view LLP through mutual contamination with identification and risk consistency [36], alongside practical adaptations to high-dimensional data [28] and adversarial training [37]. A known limitation of *pure* LLP is non-identifiability when all bags share the same proportion, as the constraints collapse to a single global mean. In contrast, NTMP supervises via *exact positive counts* in fixed/variable-size tuples and combines them with an auxiliary unlabeled pool of known prior to derive a *closed-form* unbiased risk (with stability corrections).

TABLE I
NOTATION USED THROUGHOUT THE PAPER.

Symbol	Meaning
π	Class prior $\mathbb{P}(y = +1)$
$\alpha := m/n$	Tuple-level positive ratio (tuple length n , positives m)
p_+, p_-	Class-conditionals $p(x y = \pm 1)$
p_U	Unlabeled marginal $\pi p_+ + (1 - \pi) p_-$
$\tilde{p}_T$	Per-instance marginal from tuples $\alpha p_+ + (1 - \alpha) p_-$
$g : \mathcal{X} \rightarrow \mathbb{R}$	Scoring function / classifier
$\phi(\cdot)$	Surrogate loss; $\ell(y, g) = \phi(yg)$
$\mathcal{R}(g)$	Population risk under π
ECE	Expected calibration error (bin-based; $M=15$ bins; higher is worse)
ECETS	ECE after temperature scaling (T fitted on validation by NLL)
Brier	Brier score (binary or multiclass; lower is better)
BrierTS	Brier after temperature scaling (same T as above)
$p_{\text{Holm}}(\cdot)$	Holm-adjusted p -value from paired Wilcoxon vs. URE ;
$\delta_{\text{Cliff}}(\cdot)$	liff's δ is signed so that $+$ favors URE .

Thus, while NTMP shares LLP's motivation of aggregate supervision, its identification route and training objective are closer in spirit to UU-style risk reconstruction.

III. METHOD: UNBIASED RISK + TRAINING

We present a framework for learning from n -tuples that contain exactly m positive instances (*NTMP*). We first formalize the tuple data-generation process and relate it to the underlying class-conditionals (p_+, p_-) . This yields a closed-form *unbiased risk estimator* (URE) that enables training without any instance-level labels. We then provide theory—identifiability, variance-aware weighting, and generalization—and describe a training algorithm equipped with a simple stability correction. For clarity, we develop the main results under a basic setting where both the tuple length n and the positive count m are fixed across tuples; the extensions to variable n , variable m , and jointly variable (n, m) are presented in Appendix I.

A. Data Generation

Let $D_T = \{T_i\}_{i=1}^{n_T}$ be a dataset of tuples, where $T_i = (x_i^{(1)}, \dots, x_i^{(n)})$ and each tuple contains exactly m positives and $n-m$ negatives (positions unknown). The tuple-level label space is

$$\mathcal{Y}_T = \left\{ \mathbf{y} \in \{+1, -1\}^n : \sum_{j=1}^n \mathbf{1}\{y^{(j)} = +1\} = m \right\},$$

$$|\mathcal{Y}_T| = \binom{n}{m}.$$

Lemma 1 (Tuple density as an equally weighted mixture). *Under the NTMP process, the joint density of a tuple factorizes as*

$$p_T(x^{(1)}, \dots, x^{(n)}) = \frac{1}{\binom{n}{m}} \sum_{\mathbf{y} \in \mathcal{Y}_T} \prod_{j=1}^n p(x^{(j)} | y^{(j)}),$$

where the sum is over all label assignments $\mathbf{y} = (y^{(1)}, \dots, y^{(n)})$ with exactly m entries equal to $+1$.

Flattening all tuples yields a pointwise multiset $\tilde{D}_T$ containing $n_T n$ instances. The full proof is given in Appendix B, see Lemmas A.7. The following describes its marginal.

Theorem 2 (Flattened marginal). *Let $\alpha := m/n$. If each T_i contains exactly m positives and tuples are generated independently with exchangeable positions, then the flattened index (i, j) has marginal*

$$\tilde{p}_T(x) = \alpha p_+(x) + (1 - \alpha) p_-(x),$$

so that draws sampled uniformly (with replacement) from $\tilde{D}_T$ are i.i.d. from $\tilde{p}_T$.

Thus $\tilde{p}_T$ is a fixed two-component mixture whose weights are determined by the tuple composition. The full proof is given in Appendix B, see Theorem A.8. Let $p_U(x) = \pi p_+(x) + (1 - \pi) p_-(x)$ denote the marginal of an auxiliary unlabeled dataset $D_U = \{u_i\}_{i=1}^{n_U}$, with known (or consistently estimable) prior $\pi \in (0, 1)$. Combining $\tilde{p}_T$ and p_U yields a 2×2 linear system in (p_+, p_-) .

Lemma 3 (Identification from $(\tilde{p}_T, p_U)$). *If $\alpha \neq \pi$ (identifiability), then for every x ,*

$$\begin{aligned} p_+(x) &= \frac{(1 - \alpha) p_U(x) - (1 - \pi) \tilde{p}_T(x)}{\pi - \alpha}, \\ p_-(x) &= \frac{-\alpha p_U(x) + \pi \tilde{p}_T(x)}{\pi - \alpha}. \end{aligned} \quad (\text{III.1})$$

The full proof is given in Appendix B, see Lemma A.9.

B. Unbiased Risk

Recall the supervised population risk (cf. (II.1)):

$$\mathcal{R}(g) = \pi \mathbb{E}_{p_+}[\ell(g(\mathbf{x}), +1)] + (1 - \pi) \mathbb{E}_{p_-}[\ell(g(\mathbf{x}), -1)].$$

Substituting (III.1) and using linearity of expectation expresses $\mathcal{R}(g)$ via expectations under $\tilde{p}_T$ and p_U only.

Theorem 4 (Unbiased risk representation for NTMP). *Let $\alpha = m/n$ and assume $\alpha \neq \pi$. Then for any measurable g ,*

$$\begin{aligned} \mathcal{R}(g) &= \frac{1}{\pi - \alpha} \left[\pi(1 - \alpha) \mathbb{E}_{p_U}[\ell(g(\mathbf{x}), +1)] \right. \\ &\quad - \pi(1 - \pi) \mathbb{E}_{\tilde{p}_T}[\ell(g(\mathbf{x}), +1)] \\ &\quad - (1 - \pi)\alpha \mathbb{E}_{p_U}[\ell(g(\mathbf{x}), -1)] \\ &\quad \left. + (1 - \pi)\pi \mathbb{E}_{\tilde{p}_T}[\ell(g(\mathbf{x}), -1)] \right]. \end{aligned} \quad (\text{III.2})$$

Hence $\mathcal{R}(g)$ can be estimated unbiasedly from samples of $\tilde{p}_T$ (tuples, flattened) and p_U (unlabeled pool), without any instance labels.

The full proof is given in Appendix B, see Theorem A.10.

Empirical estimator: Let $\{x_i^{(j)}\}$ be all within-tuple instances and $\{u_i\}$ the unlabeled samples. Denote the per-tuple uniform average $\frac{1}{n} \sum_{j=1}^n h(x_i^{(j)})$ to estimate $\mathbb{E}_{\tilde{p}_T}[h(\mathbf{x})]$. The plug-in empirical URE is

$$\begin{aligned} \hat{\mathcal{R}}(g) &= \frac{1}{\pi - \alpha} \left[\pi(1 - \alpha) \frac{1}{n_U} \sum_{i=1}^{n_U} \ell(g(u_i), +1) \right. \\ &\quad - \pi(1 - \pi) \frac{1}{n_T} \sum_{i=1}^{n_T} \frac{1}{n} \sum_{j=1}^n \ell(g(x_i^{(j)}), +1) \\ &\quad \left. - (1 - \pi) \frac{\alpha}{n_U} \sum_{i=1}^{n_U} \ell(g(u_i), -1) + (1 - \pi) \frac{\pi}{nn_T} \sum_{i=1}^{n_T} \sum_{j=1}^n \ell(g(x_i^{(j)}), -1) \right], \end{aligned} \quad (\text{III.3})$$

for which $\mathbb{E}[\hat{\mathcal{R}}(g)] = \mathcal{R}(g)$ holds for all g .

Variance-aware per-tuple weighting.: More generally, one could weight positions inside a tuple by nonnegative coefficients $(a_1, \dots, a_n)$ with $\sum_j a_j = 1$. The following shows the uniform choice minimizes variance.

Lemma 5 (Uniform in-tuple averaging is variance-minimizing). *Among all convex combinations $\sum_{j=1}^n a_j h(x^{(j)})$ with $a_j \geq 0$ and $\sum_j a_j = 1$, the choice $a_j = 1/n$ minimizes $\text{Var}(\sum_j a_j h(x^{(j)}))$ whenever the summands have common variance and nonnegative covariances (in particular, under i.i.d. or exchangeable symmetry within tuples). Equality holds only for the uniform weights.*

The full proof is given in Appendix B, see Lemma A.11.

Remark. We only require equal marginal variances and non-negative within-tuple covariances (i.i.d./exchangeable being special cases) for the variance-minimizing property of uniform in-tuple averaging. Independence is not required; it is used only as a convenient approximation in Appendix proofs.

A stability correction (nonnegativity clamp): Unbiased objectives in weak supervision can have high variance and take negative empirical values at finite sample sizes. We therefore use a *clamped* empirical objective

$$\widehat{\mathcal{R}}_{\text{corr}}(g) := f(\widehat{\mathcal{R}}_T(g)) + f(\widehat{\mathcal{R}}_U(g)), f(z) \in \{\text{ReLU}(z), |z|\},$$

where

$$\begin{aligned} \widehat{\mathcal{R}}_T(g) &:= \frac{1}{\pi - \alpha} \left[-\pi(1 - \pi) \widehat{\mathbb{E}}_{\tilde{p}_T}[\ell(g(\mathbf{x}), +1)] \right. \\ &\quad \left. + (1 - \pi)\pi \widehat{\mathbb{E}}_{\tilde{p}_T}[\ell(g(\mathbf{x}), -1)] \right], \\ \widehat{\mathcal{R}}_U(g) &:= \frac{1}{\pi - \alpha} \left[\pi(1 - \alpha) \widehat{\mathbb{E}}_{p_U}[\ell(g(\mathbf{x}), +1)] \right. \\ &\quad \left. - (1 - \pi)\alpha \widehat{\mathbb{E}}_{p_U}[\ell(g(\mathbf{x}), -1)] \right], \end{aligned}$$

and $\widehat{\mathbb{E}}$ denotes the corresponding empirical averages (tuples flattened uniformly for $\tilde{p}_T$). The clamp f controls variance and curbs overfitting in practice while introducing an upward bias that vanishes as sample size grows (bounded losses and concentration). Importantly, if the population components are nonnegative, the population minimizer of the original risk $\mathcal{R}(g)$ is preserved.

Remark (Identifiability and conditioning). All formulas depend on the separation factor $|\pi - \alpha|$. When α approaches π , the linear system becomes ill-conditioned and variance inflates; our robustness experiments in Section IV visualize this near-degenerate regime and confirm the theoretical prediction.

C. Class-prior Estimation and Misspecification Window

Scope. The procedure below is used *only* when the unlabeled prior is unknown or for robustness analyses in Sec. IV; our main experiments adopt the known-by-construction prior described in Sec. IV.

a) Class-prior estimation protocol: Before training on each dataset/setup, we run a three-step protocol: (i) *NP-style lower bound* to exclude pathological priors; (ii) *mixture proportion estimation (MPE)* to obtain a point estimate $\hat{\pi}$; (iii) *a controlled Δ -sweep*, where $\Delta := |\pi - \alpha|$ quantifies the misspecification between the unlabeled prior π and the tuple

mixing rate $\alpha = m/n$. This yields a reproducible operating point $\hat{\pi}$ and a data-driven robustness window $W(\Delta)$ that we report across tables/plots.

b) Three-step template with implementation options:

- 1) **NP-style lower bound (sanity gate).** Train a lightweight score model $s(\mathbf{x})$ (e.g., a 2-layer MLP or a linear probe on frozen features) to separate a small high-precision positive proxy (e.g., top- k instances from tuples with the highest predicted margins) from the unlabeled pool U . Compute the ROC curve on a held-out validation split (size: 5,000 examples or 10% of U , whichever larger). Define the NP lower bound

$$\pi_{\text{LB}} := \max_{\tau} \{ \widehat{\text{TPR}}(\tau) - \widehat{\text{FPR}}(\tau) \}_+$$

with Clopper–Pearson bands. *Options:* grid over 200 thresholds; 10-fold stratified CV if U is small.

- 2) **Point estimate via MPE.** Estimate the mixture proportion by a tail-CDF ratio with kernel smoothing:

$$\hat{\pi} = \max_{t \in \mathcal{T}} \frac{\widehat{F}_U(t)}{\widehat{F}_P(t)},$$

where $\widehat{F}(\cdot)$ are monotone tail-CDFs of $s(\mathbf{x})$ (Gaussian kernels; bandwidth by Silverman’s rule) post-processed by isotonic regression; $\mathcal{T}$ is a quantile grid (e.g., 200 points on $[0.6, 0.99]$ of P ’s score distribution). *Uncertainty:* bootstrap $B=1,000$ resamples on the validation split for a 95% percentile CI of $\hat{\pi}$.

- 3) **Δ -sweep and reporting.** Let $\Delta := |\pi - \alpha|$. Around $\hat{\pi}$, define a symmetric grid

$$\Pi = \{\hat{\pi} + \delta : \delta \in \{-0.30, -0.28, \dots, 0.30\}\} \cap (0, 1),$$

and retrain by plugging each $\pi \in \Pi$ into the URE (or reweight losses if training once with a prior head). For each $\pi \in \Pi$, evaluate the primary metric M (AP or Macro-F1) across S seeds (default $S=5$) and compute mean, std, and a 95% bootstrap CI (default $B=10,000$).

c) Recommended defaults: Validation size: 5k or 10% of U ; score model: linear head on a frozen encoder or 2-layer MLP (width 256, ReLU, dropout 0.1); kernel: Gaussian, bandwidth by Silverman; threshold grid: 200 points; bootstrap: $B=1,000$ for the $\hat{\pi}$ CI and $B=10,000$ for metric CIs; seeds: $S=5$ (reported in captions).

d) Reproducible robustness window $W(\Delta)$: Let $M(\pi)$ denote the primary metric evaluated across S seeds at prior π on the Δ -grid. We define

$$\begin{aligned} W(\Delta) = \left\{ \Delta : \underbrace{|\mathbb{E}[M(\hat{\pi})] - \mathbb{E}[M(\hat{\pi} \pm \Delta)]|}_{\text{performance drop}} \leq \varepsilon \right. \\ \left. \wedge \underbrace{\text{CIwidth}_{95\%}(M(\hat{\pi} \pm \Delta))}_{\text{stability}} \leq w^* \right\}, \end{aligned}$$

with fixed thresholds $\varepsilon=0.02$ (absolute metric points) and $w^*=0.05$ unless noted. We report the maximal contiguous interval $[\Delta_{\min}, \Delta_{\max}] \subseteq W(\Delta)$ and mark $\Delta_{\text{crit}} := \min\{\Delta : \text{CIwidth}_{95\%}(M(\hat{\pi} \pm \Delta)) \geq w^*\}$ as a red dashed line in Δ -sweep plots. Algorithm 1 summarizes the procedure.

Algorithm 1 Computing the reproducible robustness window $W(\Delta)$ (inputs: grid Π around $\hat{\pi}$, seeds S , bootstrap sizes B).

- 1: For each $\pi \in \Pi$, compute seed-wise $M(\pi)$, its mean and 95% CI (bootstrap, $B=10k$).
- 2: Let $m_0 = \mathbb{E}[M(\hat{\pi})]$. Mark $\pi \in \Pi$ as robust if $|m_0 - \mathbb{E}[M(\pi)]| \leq \varepsilon$ and $\text{CIwidth}_{95\%}(M(\pi)) \leq w^*$.
- 3: $W(\Delta) \leftarrow$ the largest contiguous set of robust values of $\Delta = |\pi - \hat{\pi}|$.
- 4: $\Delta_{\text{crit}} \leftarrow \min\{\Delta : \text{CIwidth}_{95\%}(M(\hat{\pi} \pm \Delta)) \geq w^*\}$ (draw red dashed line).

D. Conditioning and Practical Design for the $\Delta=|\pi - \alpha|$ Regime

This section complements Sec. III-C by explaining why the 2×2 identification system becomes ill-conditioned as $\alpha \rightarrow \pi$ (conditioning term $|\pi - \alpha|^{-1}$) and by fixing concrete data-design and training rules that keep the empirical URE well behaved without altering the population minimizer. Intuitively, when the tuple composition matches the unlabeled prior, the contributions from the tuple-induced and unlabeled marginals nearly cancel, amplifying noise; the variance of $\hat{\mathcal{R}}$ scales with $|\pi - \alpha|^{-2}$ up to constants, which is reflected in the generalization bound (see (III.4)). In practice, we enforce a small design margin $|\alpha - \hat{\pi}| \geq \varepsilon$ (e.g., $\varepsilon \in [0.05, 0.10]$) by mixing at least two (n, m) configurations to spread α , and—if needed—mildly reweighting/resampling the unlabeled pool so that its empirical prior stays within $\hat{\pi} \pm \varepsilon$; during training we down-weight or skip mini-batches that violate this margin and apply the nonnegativity clamp $f \in \{\text{ReLU}(\cdot), |\cdot|\}$ from Sec. III-B to reduce variance while preserving the population minimizer.

One-paragraph summary of the extensions.: For variable tuple lengths and/or counts, let $\alpha_t := m_t/n_t$ denote the per-tuple positive mixing rate and define the *effective mixing weight* $\bar{\alpha} := \mathbb{E}[\alpha_t]$ under the sampling scheme used in training (empirically, the weighted average over tuples that enter the loss). Together with the unlabeled prior π , the observable pair $(\mathbb{E}_{\text{tuple}}[\cdot], \mathbb{E}_{\text{unlab}}[\cdot])$ forms a 2×2 linear system whose unknowns are the class-conditional risks $(R_+(g), R_-(g))$. Whenever $\bar{\alpha} \neq \pi$, this system is invertible and yields the same unbiased risk estimator (URE) as in the fixed (n, m) case by replacing α with $\bar{\alpha}$. When $\bar{\alpha} = \pi$ (the ill-conditioned case), we create two strata with distinct effective rates, $\bar{\alpha}_1 \neq \bar{\alpha}_2$ (e.g., by partitioning on n_t or m_t , or by non-uniform re-sampling); solving the two 2×2 systems on each stratum restores identifiability. We then apply the same finite-sample stabilization (ABS/ReLU clamp) as in the base method. Algorithm 2 summarizes the stratify-and-solve procedure used whenever $|\hat{\alpha} - \hat{\pi}| < \tau$.

Theorem 6 (Identifiability with heterogeneous tuples). *Let $\alpha_t = m_t/n_t$ be the tuple-level positive mixing rate and $\bar{\alpha} := \mathbb{E}[\alpha_t]$ the effective rate induced by the sampling scheme (empirically, the average over tuples used by the loss). Let π be the class prior of the unlabeled pool. Consider the linear*

Algorithm 2 Stratify-and-solve for NTMP under $\hat{\alpha} \approx \hat{\pi}$. Partition tuples to induce $\hat{\alpha}_1 \neq \hat{\alpha}_2$, solve two 2×2 URE systems, aggregate, and apply ABS/ReLU stabilization.

- 1: Compute $\hat{\alpha}_t \leftarrow m_t/n_t$, $\hat{\alpha} \leftarrow \text{Avg}(\hat{\alpha}_t)$.
- 2: **if** $|\hat{\alpha} - \hat{\pi}| \geq \tau$ **then**
- 3: Use $\hat{\alpha}$ in the 2×2 system and train.
- 4: **else**
- 5: Partition tuples to obtain $\hat{\alpha}_1 \neq \hat{\alpha}_2$; solve per stratum and aggregate.
- 6: **end if**
- 7: Apply ABS/ReLU clamp and optimize g .

system linking observables to class-conditional risks:

$$\begin{bmatrix} \bar{\alpha} & 1 - \bar{\alpha} \\ \pi & 1 - \pi \end{bmatrix} \begin{bmatrix} R_+(g) \\ R_-(g) \end{bmatrix} = \begin{bmatrix} T(g) \\ U(g) \end{bmatrix},$$

$$T(g) := \mathbb{E}_{\text{tuple}}[\ell(g)], \quad U(g) := \mathbb{E}_{\text{unlab}}[\ell(g)].$$

(i) If $\bar{\alpha} \neq \pi$, the system is invertible and the unbiased risk estimator is obtained by replacing the fixed- α coefficients with $\bar{\alpha}$. (ii) If $\bar{\alpha} = \pi$, identifiability is restored by constructing two strata $\mathcal{S}_1, \mathcal{S}_2$ with distinct effective rates $\bar{\alpha}_1 \neq \bar{\alpha}_2$ (e.g., partition on n_t or m_t , or apply non-uniform re-sampling), and solving the two systems on $\mathcal{S}_1$ and $\mathcal{S}_2$ separately. (iii) In practice we use the same finite-sample stabilization as the base method (ABS/ReLU clamp) after recovering (R_+, R_-) .

E. Theoretical Analysis

We present generalization guarantees for the NTMP learner, show consistency with optimal rates, and describe a simple stability correction. Throughout this subsection we assume the identifiability condition $\alpha \neq \pi$ and use the unbiased risk representation in (III.2) with its empirical counterpart (III.3).

Complexity measure.: For a function class $\mathcal{G} \subset \{g : \mathcal{X} \rightarrow \mathbb{R}\}$ and a distribution μ on $\mathcal{X}$, we write the (distribution-dependent) Rademacher complexity with s i.i.d. samples $X_{1:s} \sim \mu$ as

$$\mathfrak{R}_s^\mu(\mathcal{G}) := \mathbb{E}_{X_{1:s}} \mathbb{E}_{\sigma_{1:s}} \left[\sup_{g \in \mathcal{G}} \frac{1}{s} \sum_{i=1}^s \sigma_i g(X_i) \right],$$

where $\sigma_i \stackrel{\text{i.i.d.}}{\sim} \text{Unif}\{\pm 1\}$. Assume $\phi(t) := \ell(t, +1)$ and $\psi(t) := \ell(t, -1)$ are ρ -Lipschitz in t and that $|g(\mathbf{x})| \leq C_g$ for all $g \in \mathcal{G}, \mathbf{x} \in \mathcal{X}$, which implies bounded losses $0 \leq \ell(g(\mathbf{x}), y) \leq C_\ell$ with $C_\ell := \max\{\phi(C_g), \psi(C_g)\}$.

Theorem 7 (Generalization bound and rate). *Let $g^* \in \arg \min_{g \in \mathcal{G}} \mathcal{R}(g)$ and $\hat{g} \in \arg \min_{g \in \mathcal{G}} \hat{\mathcal{R}}(g)$, where $\hat{\mathcal{R}}$ is the empirical URE in (III.3) based on n_T tuples (each of length n) and n_U unlabeled points. Then, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\begin{aligned} \mathcal{R}(\hat{g}) - \mathcal{R}(g^*) &\leq \frac{C_1 \rho}{|\pi - \alpha|} (\mathfrak{R}_{n_T n}^{\tilde{p}_T}(\mathcal{G}) + \mathfrak{R}_{n_U}^{p_U}(\mathcal{G})) \\ &\quad + \frac{C_2 C_\ell}{|\pi - \alpha|} \left(\sqrt{\frac{\log \frac{2}{\delta}}{n_T n}} + \sqrt{\frac{\log \frac{2}{\delta}}{n_U}} \right), \end{aligned} \quad (\text{III.4})$$

for absolute constants $C_1, C_2 > 0$. In particular, if $\mathfrak{R}_s^\mu(\mathcal{G}) \leq C_G/\sqrt{s}$, then

$$\mathcal{R}(\hat{g}) - \mathcal{R}(g^*) = \mathcal{O}\left(\frac{1}{|\pi - \alpha|}\right) \cdot \left(\frac{1}{\sqrt{n_T n}} + \frac{1}{\sqrt{n_U}}\right) \quad (\text{ignoring log factors}).$$

Thus the estimator is consistent as $n_T, n_U \rightarrow \infty$ and achieves the optimal $N^{-1/2}$ rate with $N \approx n_T n + n_U$ effective samples.

Scaling near $\Delta = |\pi - \alpha|$. At the population-variance level, the matrix inversion amplifies noise at rate $\propto \Delta^{-2}$; correspondingly, the estimation-error terms in the generalization bound (Eq. (III.4)) scale as $\mathcal{O}(\Delta^{-1})$. This variance-vs.-bound distinction matches the empirical “ill-conditioned band” in Fig. 2.

Proof sketch. Apply symmetrization to (III.3) and (III.2) separately for the $\tilde{p}_T$ and p_U parts, use the contraction lemma to pass from g to the Lipschitz losses $\ell(g, \pm 1)$, and combine deviations via a union bound. The coefficients in (III.2) contribute the factor $|\pi - \alpha|^{-1}$, and standard Hoeffding-type bounds for bounded losses yield the concentration terms. Full details are deferred to an accompanying Appendix B, see Theorem A.16. $\square$

Variance-optimal in-tuple averaging. As noted after (III.3), replacing the in-tuple uniform average by $\sum_{j=1}^n a_j h(x^{(j)})$ with $\sum_j a_j = 1$ increases variance unless $a_j \equiv 1/n$ (Lemma 5). We therefore adopt uniform in-tuple averaging throughout to tighten the constants in (III.4).

Remarks. (i) The factor $|\pi - \alpha|^{-1}$ in (III.4) captures the conditioning of the 2×2 system; as $\alpha \rightarrow \pi$ the problem becomes ill-conditioned, matching the robustness study in Section IV. (ii) If ϕ, ψ are L -Lipschitz and $|g| \leq G$, one may replace C_ℓ by LG in (III.4). (iii) The bound extends to variable (n, m) by conditioning on strata of α and averaging the corresponding deviations. (iv) *Practical tip.* To avoid the near-degenerate regime $\alpha \approx \pi$, enforce a design margin $|\alpha - \hat{\pi}| \geq \varepsilon$ (e.g., $\varepsilon \in [0.05, 0.10]$): mix tuples from at least two (n, m) configurations to spread α , and, if needed, mildly reweight/resample the unlabeled pool so that its empirical prior stays within $\hat{\pi} \pm \varepsilon$. During training, down-weight or skip mini-batches whose current α violates this margin to keep the URE well-conditioned.

F. Risk Bias and Excess Risk under Prior Misspecification

In practice, the true class prior π is rarely known and is replaced by an estimate $\hat{\pi}$ from side information or a preliminary model. Because identification of (p_+, p_-) from $(p_U, \tilde{p}_T)$ relies on the 2×2 system with determinant $(\pi - \alpha)$, any prior error may be amplified when $\alpha = \frac{m}{n}$ is close to π . This subsection quantifies how such misspecification propagates to the population risk. We assume a bounded surrogate ϕ (e.g., logistic, ramp, hinge clipped to a finite range) and emphasize the identifiability condition $\alpha \neq \pi$.

Theorem 8 (Risk bias under misspecified class prior). *Let $\pi := \mathbb{P}(y = +1) \in (0, 1)$ and let $p_U = \pi p_+ + (1 - \pi) p_-$. In*

the NTMP setting with tuple length n and positive count m , the flattened marginal is

$$\tilde{p}_T = \alpha p_+ + (1 - \alpha) p_-, \quad \alpha := \frac{m}{n},$$

and $\alpha \neq \pi$. Let $\ell(y, g(\mathbf{x})) = \phi(y g(\mathbf{x}))$ with $0 \leq \phi \leq B$, and define

$$\mathcal{R}(g) = \pi \mathbb{E}_{p_+}[\phi(g(\mathbf{x}))] + (1 - \pi) \mathbb{E}_{p_-}[\phi(-g(\mathbf{x}))].$$

Consider the plug-in representation using a misspecified prior $\hat{\pi} = \pi + \delta \in (0, 1)$ with $\hat{\pi} \neq \alpha$:

$$\hat{p}_+^{(\hat{\pi})} = \frac{(1 - \alpha) p_U - (1 - \hat{\pi}) \tilde{p}_T}{\hat{\pi} - \alpha}, \quad \hat{p}_-^{(\hat{\pi})} = \frac{-\alpha p_U + \hat{\pi} \tilde{p}_T}{\hat{\pi} - \alpha},$$

and the plug-in population risk

$$\tilde{\mathcal{R}}(g; \hat{\pi}) = \pi \mathbb{E}_{\hat{p}_+^{(\hat{\pi})}}[\phi(g(\mathbf{x}))] + (1 - \pi) \mathbb{E}_{\hat{p}_-^{(\hat{\pi})}}[\phi(-g(\mathbf{x}))].$$

Then for every measurable g ,

$$|\tilde{\mathcal{R}}(g; \hat{\pi}) - \mathcal{R}(g)| \leq \frac{2B|\delta|}{\gamma^2}, \quad \gamma := \min_{\xi \in \{\min\{\pi, \hat{\pi}\}, \max\{\pi, \hat{\pi}\}\}} |\xi - \alpha|.$$

In particular, if π and $\hat{\pi}$ lie on the same side of α , then $\gamma = \min\{|\pi - \alpha|, |\hat{\pi} - \alpha|\}$ and

$$|\tilde{\mathcal{R}}(g; \hat{\pi}) - \mathcal{R}(g)| \leq \frac{2B|\delta|}{\min\{|\pi - \alpha|, |\hat{\pi} - \alpha|\}^2}.$$

Moreover, let $g^ \in \arg \min_{g \in \mathcal{G}} \mathcal{R}(g)$ and $\tilde{g} \in \arg \min_{g \in \mathcal{G}} \tilde{\mathcal{R}}(g; \hat{\pi})$. Then*

$$\mathcal{R}(\tilde{g}) - \mathcal{R}(g^*) \leq 2 \sup_{g \in \mathcal{G}} |\tilde{\mathcal{R}}(g; \hat{\pi}) - \mathcal{R}(g)| \leq \frac{4B|\delta|}{\gamma^2}.$$

The full proof is given in Appendix B, see Theorem A.14. **Takeaways.** (i) *Conditioning effect:* the factor γ^{-2} captures the ill-conditioning as $|\pi - \alpha| \rightarrow 0$, explaining sensitivity to prior errors. (ii) *Uniform control:* for fixed α , the bias scales linearly in $|\delta|$ and is uniformly bounded across g for bounded ϕ .

Guidelines. Design side: avoid regimes where α concentrates too closely around π (e.g., mix tuples with distinct (n, m) so that α varies). Estimation side: prefer prior estimators with small absolute error and, when possible, shrink $\hat{\pi}$ away from α if they fall on opposite sides. Mitigation: if ϕ is L -Lipschitz and $|g(\mathbf{x})| \leq G$ a.s., replace B by LG in the above bounds.

G. Misspecified Tuple Count (Random m)

The positive count m may be noisy due to annotation errors or stochastic tuple construction; randomness in m translates into randomness in $\alpha = m/n$, yielding a *randomly conditioned* inversion. We model the practitioner’s input as a random $\hat{m} \sim p(m')$ and study how the plug-in risk deviates from the truth.

Theorem 9 (Risk bias under random count misspecification). *Let $\mathbf{x} \in \mathcal{X}$, $y \in \{+1, -1\}$ with prior $\pi \in (0, 1)$, class-conditionals p_+, p_- , and unlabeled marginal $p_U = \pi p_+ + (1 - \pi) p_-$. For tuples of length n with true count m and $\alpha = m/n$,*

$$\tilde{p}_T = \alpha p_+ + (1 - \alpha) p_-, \quad \text{assume } \pi \neq \alpha.$$

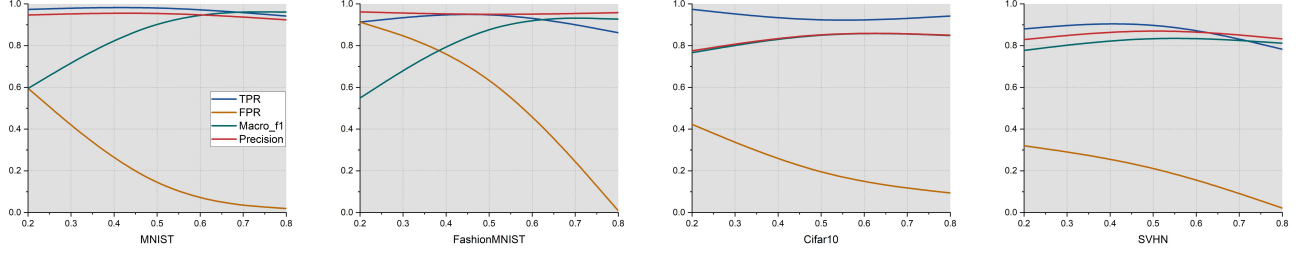


Fig. 1. Tuples of length $n=3$ with $m=1$. Horizontal axis: class prior π ; vertical axis: metric value. As π increases, TPR , $Macro-F_1$, and $Precision$ remain stable or improve across datasets, while FPR decreases markedly, indicating that prior shifts predominantly influence false positives.

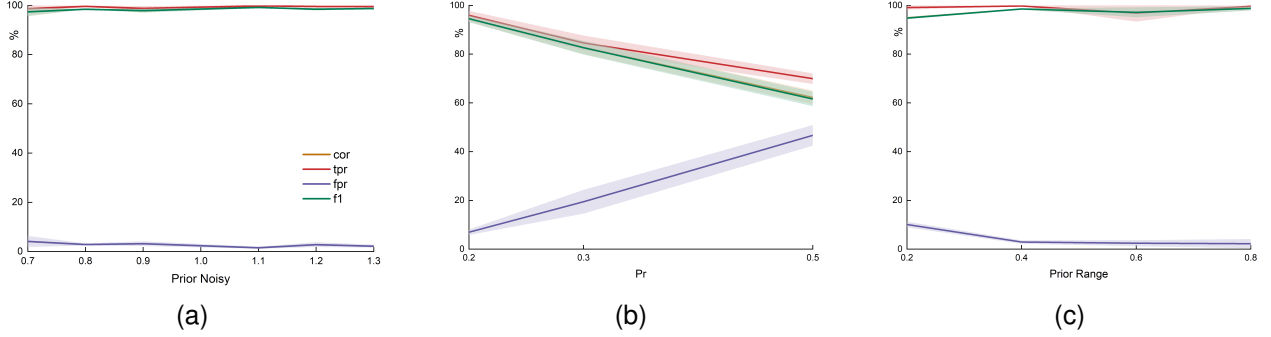


Fig. 2. Robustness on MNIST (three repeats; solid lines = mean; shaded = $\pm$ one std). We report **cor**, **TPR**, **FPR**, **F1**. (a) *Class-prior misspecification*: perturbing the assumed prior within $\pm 30\%$ barely changes **cor/F1** and keeps **FPR** near zero. (b) *Tuple-count noise*: randomly increasing/decreasing m by one with probability Pr raises **FPR** roughly linearly and monotonically reduces **cor/TPR/F1**. (c) *Ill-conditioned regime*: sweeping π across a range that includes the near-degenerate region shows largely stable metrics with small fluctuations and consistently low **FPR**, except in the immediate vicinity of the ill-conditioned boundary.

Let $\ell(y, g(\mathbf{x})) = \phi(yg(\mathbf{x}))$ with $0 \leq \phi \leq B$ and define $\mathcal{R}(g) = \pi \mathbb{E}_{p_+}[\phi(g(\mathbf{x}))] + (1 - \pi) \mathbb{E}_{p_-}[\phi(-g(\mathbf{x}))]$. Suppose, instead of α , we use a random $\hat{\alpha} := \hat{m}/n$ with $\hat{m} \sim p(m')$ and $\hat{\alpha} \neq \pi$ a.s. Given $(p_U, \tilde{p}_T)$, form

$$\hat{p}_+^{(\hat{\alpha})} = \frac{(1 - \hat{\alpha})p_U - (1 - \pi)\tilde{p}_T}{\pi - \hat{\alpha}}, \quad \hat{p}_-^{(\hat{\alpha})} = \frac{-\hat{\alpha}p_U + \pi\tilde{p}_T}{\pi - \hat{\alpha}},$$

and the plug-in risk $\tilde{\mathcal{R}}(g; \hat{\alpha})$ accordingly. Then, for every g and every realization of $\hat{\alpha}$,

$$|\tilde{\mathcal{R}}(g; \hat{\alpha}) - \mathcal{R}(g)| \leq \frac{2B|\hat{\alpha} - \alpha|}{\eta^2},$$

$$\eta := \min_{\zeta \in [\min\{\alpha, \hat{\alpha}\}, \max\{\alpha, \hat{\alpha}\}]} |\pi - \zeta|.$$

Taking expectation over $\hat{\alpha}$,

$$\mathbb{E}_{\hat{\alpha}}[|\tilde{\mathcal{R}}(g; \hat{\alpha}) - \mathcal{R}(g)|] \leq \frac{2B}{\underline{\eta}^2} \mathbb{E}_{\hat{\alpha}}[|\hat{\alpha} - \alpha|], \quad \underline{\eta} := \inf_{\zeta \in \mathcal{I}} |\pi - \zeta|,$$

where $\mathcal{I}$ contains (a.s.) the supports of α and $\hat{\alpha}$. Moreover, if $\bar{\mathcal{R}}(g) := \mathbb{E}_{\hat{\alpha}}[\tilde{\mathcal{R}}(g; \hat{\alpha})]$ and $\bar{g} \in \arg \min_{g \in \mathcal{G}} \bar{\mathcal{R}}(g)$, then

$$\mathcal{R}(\bar{g}) - \mathcal{R}(g^*) \leq 2 \sup_{g \in \mathcal{G}} |\bar{\mathcal{R}}(g) - \mathcal{R}(g)| \leq \frac{4B}{\underline{\eta}^2} \mathbb{E}_{\hat{\alpha}}[|\hat{\alpha} - \alpha|].$$

The full proof is given in Appendix B, see Theorem A.15.

Corollary 10 (Variance-form and high-probability variants). Let $\sigma_\alpha^2 := \mathbb{E}[(\hat{\alpha} - \alpha)^2]$. Then $\mathbb{E}|\hat{\alpha} - \alpha| \leq \sigma_\alpha$ and thus

$$\sup_g |\bar{\mathcal{R}}(g) - \mathcal{R}(g)| \leq \frac{2B}{\underline{\eta}^2} \sigma_\alpha, \quad \mathcal{R}(\bar{g}) - \mathcal{R}(g^*) \leq \frac{4B}{\underline{\eta}^2} \sigma_\alpha.$$

If $|\hat{\alpha} - \alpha| \leq \varepsilon$ a.s., then $|\tilde{\mathcal{R}}(g; \hat{\alpha}) - \mathcal{R}(g)| \leq \frac{2B\varepsilon}{\underline{\eta}^2}$ for all g . If $\hat{\alpha} - \alpha$ is sub-Gaussian with proxy variance v^2 , then with probability at least $1 - \delta$,

$$|\tilde{\mathcal{R}}(g; \hat{\alpha}) - \mathcal{R}(g)| \leq \frac{2B}{\underline{\eta}^2} v \sqrt{2 \log(1/\delta)}.$$

Remarks. (i) The conditioning factor is $|\pi - \hat{\alpha}|^{-2}$, mirroring Theorem 8; the system is ill-conditioned as $\hat{\alpha} \rightarrow \pi$. (ii) Mixing tuples with diverse positive ratios enlarges $\underline{\eta}$, improving robustness. (iii) If ϕ is L -Lipschitz and $|g(\mathbf{x})| \leq G$ a.s., replace B by LG throughout.

IV. EXPERIMENTS

We empirically evaluate the proposed NTMP framework on four image benchmarks and compare it against relevant baselines. Our study addresses three questions: (i) Does NTMP achieve higher accuracy than baselines under count-only supervision? (ii) Do the stability corrections (ABS/ReLU clamps) mitigate overfitting? (iii) How do the class prior π and the tuple configuration (n, m) affect COR/TPR/FPR/ F_1 (and threshold-free metrics such as AP/AUROC)?

A. Experimental Setup

Default prior regime and variants. Unless otherwise stated, we adopt the **known-by-construction** prior regime: the unlabeled pool D_U is obtained by stripping labels from a disjoint training split, so its class prior π is known and plugged

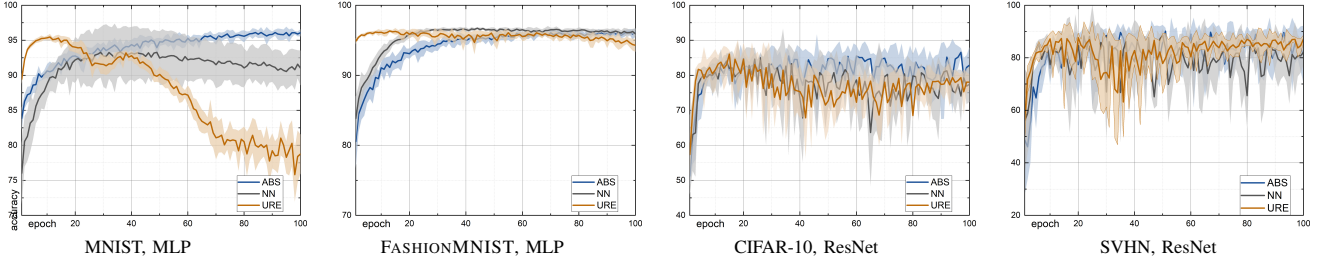


Fig. 3. Class prior $\pi=0.2$, tuples $(n, m) = (3, 1)$. Horizontal axis: training iterations; vertical axis: accuracy. Mean $\pm$ std over five trials. The plain URE may overfit (accuracy spikes then decays), whereas the corrected variants stabilize training and improve the final accuracy (see text).

into the URE. For robustness and the **unknown-prior** variant, we instead estimate a prior $\hat{\pi}$ following Sec. III-C (NP lower bound $\rightarrow$ MPE point estimate $\rightarrow \Delta$ -sweep), and we report the operating gap $\Delta = |\alpha - \hat{\pi}|$ and the reproducible window $W(\Delta)$.

Datasets and binary remappings.: Each dataset is converted to a binary task and then used to construct NTMP tuples. We report the specific (n, m) (and the implied $\alpha = m/n$) in the captions of the corresponding tables/figures.

- **MNIST** [38]: 28×28 grayscale digits (60k/10k train/test). *Positive* = even digits $\{0, 2, 4, 6, 8\}$; *negative* = odd digits $\{1, 3, 5, 7, 9\}$ (balanced $\pi=0.5$).
- **FASHIONMNIST** [39]: 28×28 grayscale clothes (60k/10k). *Positive* = {T-shirt/top, Pullover, Coat, Shirt, Bag}; *negative* = {Trouser, Dress, Sandal, Sneaker, Ankle boot} ($\pi=0.5$).
- **SVHN** [40]: 32×32 color digits (73,257/26,032). *Positive* = even digits; *negative* = odd digits (consistent with MNIST; $\pi=0.5$).
- **CIFAR-10** [41]: 32×32 color images (50k/10k). *Positive* = {airplane, bird, cat, dog, horse, truck}; *negative* = {automobile, deer, frog, ship} (slightly imbalanced $\pi=0.6$).

Images are scaled to $[0, 1]$. No augmentation is used unless specified.

Tuple construction and unlabeled pool. For each dataset, we construct D_T by sampling n_T tuples of fixed length $n \in \{3, 5, 7\}$ with an exact positive count $m \in \{1, 2\}$ (so $\alpha = m/n$). Each tuple $T_i = (x_i^{(1)}, \dots, x_i^{(n)})$ is formed by drawing instances *without replacement* from the training split subject to the count constraint. We choose n_T so that $n_T n$ matches the original training size (e.g., for MNIST, $n_T = 60,000/n$). The auxiliary unlabeled set D_U is created by stripping labels from a disjoint training pool; its class prior π is therefore *known by construction* and is used in the URE terms (Section III). For robustness analyses (prior-misspecification sweeps), we follow the protocol in Section III-C and report S -seed means with 95% bootstrap CIs (captions specify S). Under the unknown-prior variant, we estimate $\hat{\pi}$ using the pipeline in Section III-C (NP lower bound $\rightarrow$ MPE $\rightarrow \Delta$ -sweep) and then plug $\hat{\pi}$ into the same URE; unless otherwise stated, $\hat{\pi}$ is fixed per setup.

Baselines and metrics.: We compare against representative weak-supervision baselines (UU and clustering-style surrogates), trained on the same backbone and data splits. Primary metrics are Macro- F_1 , AP, and AUROC; we additionally report COR/TPR/FPR to diagnose error modes. All reported

TABLE II
SCOPE, IDENTIFIABILITY, AND TYPICAL FAILURE MODES ACROSS PARADIGMS. NTMP EXPLOITS EXACT TUPLE COUNTS; LLP RELIES ON CROSS-BAG PROPORTION VARIATION; MIL LEVERAGES INTRA-BAG STRUCTURE AND INSTANCE DEPENDENCIES.

Paradigm	When it shines	Typical failure mode
NTMP	Exact tuple counts (n, m) available; orderless tuples; unlabeled prior π given/estimable; $\bar{\alpha} \neq \pi$ so the 2×2 system is invertible.	Near-ill-conditioned regime with $\bar{\alpha} \approx \pi$ and no feasible stratification (insufficient data or distributional shift); strong intra-tuple structure that is lost under mixture flattening.
LLP	Only bag-level proportions (no exact counts); sufficient cross-bag variation of proportions; many bags per class mixture.	Bags have nearly identical proportions $\Rightarrow$ non-identifiable or ill-posed; high variance in small-sample settings.
MIL	Strong intra-bag spatial/temporal structure or instance interactions (locality, co-occurrence, ordering); “at-least-one” semantics are natural.	Instances are approximately i.i.d. within bags and structure is weak; attention may overfit to spurious patterns; under pure count supervision, the model may lock onto wrong salient regions.

numbers are across S seeds (default $S=5$) with 95% bootstrap CIs (default $B=10,000$), matching the robustness protocol in Section III-C.

Training and stability corrections.: Models are trained by minimizing the empirical URE in (III.3) with mini-batches drawn from D_T and D_U . As predicted by the theory (conditioning factor $|\pi - \alpha|^{-1}$ in (III.4)), variance inflates near $\alpha \approx \pi$. We therefore report results both with and without the nonnegative clamp (ABS/ReLU) described in Section III-B; the clamp stabilizes finite-sample behavior without altering the population optimum of $\mathcal{R}(g)$.

Reporting of (n, m) and prior sweeps.: To connect experiments to the misspecification theory (Sections III-F–III-G), we accompany main tables/plots with (i) the operating gap $\Delta = |\alpha - \hat{\pi}|$, (ii) Δ -sweep curves (when applicable) with Δ_{crit} marked, and (iii) explicit (n, m) for each result.

Metrics and statistical tests. For the main metrics we report AP (average precision), AUROC, and F_1 . Calibration

TABLE III

MAIN RESULTS ON FASHIONMNIST ($S=5$ SEEDS). MEAN $\pm$ STD; 95% CI IN BRACKETS. p -VALUES ARE HOLM-ADJUSTED PAIRED WILCOXON VS. URE; δ_{Cliff} IS CLIFF'S δ .

Δ	Method	AP $\uparrow$	AUROC $\uparrow$	ECE _{TS} $\downarrow$	Brier _{TS} $\downarrow$	$p_{\text{Holm}}(\text{AP})$	Cliff's $\delta(\text{AP})$
0.00	ABS	0.99 $\pm$ 0.008[0.002]	0.99 $\pm$ 0.008[0.002]	0.11[0.005]	0.06[0.012]	0.000	0.815
0.00	URE	0.77 $\pm$ 0.279[0.086]	0.69 $\pm$ 0.382[0.117]	0.29[0.090]	0.27[0.090]	–	–
0.25	ABS	0.99 $\pm$ 0.005[0.002]	0.99 $\pm$ 0.003 [0.001]	0.09[0.007]	0.05[0.008]	0.000	0.331
0.25	URE	0.36 $\pm$ 0.033[0.019]	0.14 $\pm$ 0.142[0.073]	0.59[0.120]	0.52[0.127]	–	–

metrics include the bin-based expected calibration error

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|,$$

where $B_m = \{i : \hat{p}_i \in ((m-1)/M, m/M]\}$, $\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbb{I}\{\hat{y}_i = y_i\}$, $\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \hat{p}_i$, N is the number of instances, and M the number of bins (default $M=15$). The Brier score is

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - \mathbb{I}\{y_i=1\})^2 \quad (\text{binary});$$

$$\text{Brier}_{\text{mc}} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K (\hat{p}_{ik} - \mathbb{I}\{y_i=k\})^2 \quad (\text{multiclass}).$$

We additionally report temperature-scaled variants ECETS and BrierTS, where scores are calibrated by a scalar temperature $T>0$ fitted on a held-out validation set by minimizing the negative log-likelihood (softmax/sigmoid with logits divided by T), and then applied to the test set.

For significance, we use a paired Wilcoxon signed-rank test comparing each method to OURS across identical seeds/splits and adjust the resulting p -values by Holm's step-down procedure; we denote the adjusted value by p_{Holm} and write $p_{\text{Holm}}(\text{AP})$, $p_{\text{Holm}}(\text{AUROC})$, etc. Effect size is summarized by Cliff's delta on the paired differences:

$$\delta_{\text{Cliff}} = \frac{\#\{(i, j) : x_i > y_j\} - \#\{(i, j) : x_i < y_j\}}{n_x n_y},$$

where $\{x_i\}$ and $\{y_j\}$ are the metric values of OURS and the comparator under the same runs (for paired designs, use $(x_i - y_i)$'s induced ordering). We follow the common interpretation thresholds: $|\delta_{\text{Cliff}}| < 0.147$ negligible, < 0.33 small, < 0.474 medium, otherwise large. Unless noted, all curves/tables show mean $\pm$ 95% bootstrap CI over seeds.

B. Models and Training Protocol

MNIST/FASHIONMNIST. A 4-layer MLP (300 units per layer, ReLU, BatchNorm [42]) producing a scalar score $g(\mathbf{x})$. Optimizer: Adam [43], learning rate 10^{-4} , 100 epochs.

SVHN/CIFAR-10. A lightweight ResNet-20 (9 residual blocks), Adam with learning rate 5×10^{-4} , 100 epochs; standard per-channel normalization. No heavy augmentation is used to isolate supervision effects.

Optimization of URE. We minimize the empirical unbiased risk (III.3) over mixed mini-batches (tuples from D_T and samples from D_U). Unless otherwise noted, we use *uniform in-tuple averaging* (Lemma 5) and optionally apply

a *nonnegative clamp* (ABS/ReLU) inside the tuple/unlabeled terms (Section III-B) to stabilize training. As predicted by the conditioning factor $|\pi - \alpha|^{-1}$, we additionally report results both with and without clamping when $\alpha \approx \pi$.

C. Baselines

We compare NTMP against methods that operate under weak/aggregate supervision.

a) *K-Means (KM)*: Unsupervised clustering with $k=2$ on raw (or penultimate) features. Clusters are mapped to labels by matching cluster proportions to the known π via a two-way assignment (equivalent to majority mapping when $\pi=0.5$). This provides a purely unsupervised reference.

b) *UU learning*: Risk reconstruction from two unlabeled sets with distinct priors [23], [24]. We instantiate one set as D_U (prior π) and the other as the flattened tuple pool $\tilde{D}_T$ (effective prior α), yielding a direct UU objective on the same backbone. UUCor denotes UU unbiased risk with absolute corrected function.

c) *Clustering+Classifier*: Cluster first, then train a linear classifier on cluster pseudo-labels (self-training style). This probes whether count signals in NTMP offer gains beyond structure-only assignments.

d) *Attention-MIL (non-LLM)*: A representative MIL baseline with attention pooling [30]. Each bag (tuple) $\mathcal{B} = \{x_i\}_{i=1}^L$ is encoded by a *frozen* image encoder into $z_i \in \mathbb{R}^d$; an attention head produces $a_i = \text{softmax}_i(w^\top \tanh(Wz_i))$, and the bag embedding $h = \sum_i a_i \tanh(Wz_i)$ feeds a linear classifier $s_{\text{bag}} = v^\top h$. We train only the attention and linear heads with a bag-level binary loss, using identical backbones, budgets, and seeds as NTMP. For instance-level analysis, we also report per-instance margins $s_i = v^\top \tanh(Wz_i)$ without instance labels.

e) *CLIP-MIL (LLM-assisted)*: A lightweight LLM-assisted MIL variant using a *frozen* CLIP encoder. Let $f(\cdot)$ and $g(\cdot)$ be CLIP image/text encoders, with normalized embeddings $\hat{f}(x)$ and $\hat{g}(t)$. The instance margin is the cosine score $s_i = \langle \hat{f}(x_i), \hat{g}(t) \rangle$ for a class prompt; the bag score uses a permutation-invariant aggregator (mean pooling): $s_{\text{bag}} = \frac{1}{L} \sum_i s_i$. No additional head is fitted; supervision remains strictly at the bag level.

f) *Bag-level Cross-Entropy Proportion Matching (LLP)*: For each bag, average instance softmax outputs to obtain $\bar{\mathbf{p}}$ and match to the target proportions $\mathbf{q}$ via $H(\mathbf{q}, \bar{\mathbf{p}})$; a light instance-entropy penalty stabilizes training. This is a standard, strong LLP baseline.

TABLE IV
ACCURACY ON FOUR DATASETS WITH $(n, m) = (3, 1)$ AND CLASS PRIOR $\pi \in \{0.2, 0.5, 0.8\}$. VALUES ARE MEAN $\pm$ STD OVER FIVE RUNS. ABS/ReLU DENOTE NONNEGATIVE CLAMPS APPLIED INSIDE THE TUPLE AND UNLABELED TERMS OF THE URE.

prior	Datasets	P_{NTMPU}	Baseline					
		ABS	NN	URE	UUcor	UU	KM	KM++
0.2	MNIST	95.50 $\pm$ 0.09	96.44$\pm$0.05	94.90 $\pm$ 0.25	92.94 $\pm$ 0.32	85.92 $\pm$ 0.81	58.44 $\pm$ 0.03	58.52 $\pm$ 0.01
	FASHIONMNIST	93.08 $\pm$ 0.33	93.57$\pm$0.20	84.51 $\pm$ 0.54	92.60 $\pm$ 0.12	91.30 $\pm$ 0.44	73.11 $\pm$ 0.01	73.15 $\pm$ 0.01
	CIFAR10	80.94$\pm$1.49	72.20 $\pm$ 3.07	73.29 $\pm$ 1.56	76.08 $\pm$ 0.45	69.37 $\pm$ 1.60	50.80 $\pm$ 0.00	50.63 $\pm$ 0.00
	SVHN	79.77 $\pm$ 1.80	87.15 $\pm$ 1.45	87.57$\pm$0.84	71.92 $\pm$ 2.37	82.82 $\pm$ 0.46	52.34 $\pm$ 0.00	52.14 $\pm$ 0.00
0.5	MNIST	97.76 $\pm$ 0.44	98.87$\pm$0.18	87.53 $\pm$ 0.46	68.84 $\pm$ 0.61	61.88 $\pm$ 0.33	60.73 $\pm$ 0.01	59.96 $\pm$ 0.01
	FASHIONMNIST	93.45 $\pm$ 0.67	96.39$\pm$0.55	78.50 $\pm$ 0.73	73.83 $\pm$ 1.03	73.01 $\pm$ 0.00	73.48 $\pm$ 0.00	73.42 $\pm$ 0.00
	CIFAR10	80.01$\pm$2.11	73.99 $\pm$ 2.61	69.17 $\pm$ 1.50	67.80 $\pm$ 1.39	62.24 $\pm$ 0.77	51.12 $\pm$ 0.01	51.04 $\pm$ 0.00
	SVHN	78.11 $\pm$ 1.83	86.12$\pm$1.91	83.66 $\pm$ 0.72	59.90 $\pm$ 2.29	50.22 $\pm$ 5.85	51.83 $\pm$ 0.00	51.84 $\pm$ 0.00
0.8	MNIST	98.98 $\pm$ 0.17	99.07$\pm$0.20	94.88 $\pm$ 0.25	87.25 $\pm$ 0.51	73.90 $\pm$ 0.14	51.33 $\pm$ 0.01	51.41 $\pm$ 0.01
	FASHIONMNIST	93.47 $\pm$ 0.74	96.21$\pm$0.33	82.20 $\pm$ 0.65	89.77 $\pm$ 1.28	73.27 $\pm$ 0.10	69.72 $\pm$ 0.04	70.84 $\pm$ 0.03
	CIFAR10	79.43$\pm$1.27	73.90 $\pm$ 1.91	66.91 $\pm$ 2.22	74.21 $\pm$ 0.93	69.03 $\pm$ 0.56	50.29 $\pm$ 0.01	50.34 $\pm$ 0.03
	SVHN	84.25$\pm$2.52	80.65 $\pm$ 2.04	65.61 $\pm$ 3.65	77.98 $\pm$ 3.50	62.56 $\pm$ 8.96	53.75 $\pm$ 0.00	54.07 $\pm$ 0.00

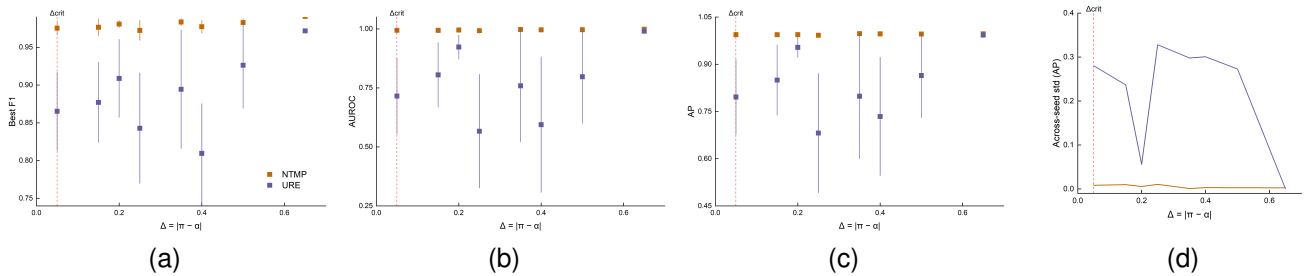


Fig. 4. **Metrics and stability under prior misspecification on MNIST.** Horizontal axis: $\Delta = |\pi - \alpha|$, where π is the training prior and $\alpha = m/n$ is the tuple positive rate for the (n, m) configuration. We compare **URE+ABS** with **URE**. Markers show the mean over $S=3$ seeds; vertical bars denote 95% bootstrap confidence intervals. The red dashed line marks Δ_{crit} , defined as the smallest gap at which the 95% CI width of AP first exceeds the preset threshold $w^*=0.05$ (cf. Section III-C). (a) Best- F_1 vs. Δ . (b) AUROC vs. Δ . (c) AP vs. Δ . (d) Stability vs. Δ , quantified by the across-seed standard deviation of AP (higher is less stable). Across the sweep, NTMP maintains high accuracy with near-zero dispersion, whereas URE degrades and becomes increasingly variable as misspecification grows. *Settings:* $n \in \{5, 10\}$, $m \in \{1, 2, 3, 4\}$, $\pi \in \{0.2, 0.4, 0.6, 0.8\}$.

TABLE V
SMALL-SCALE COMPARISON ON MNIST TUPLES UNDER COUNT-ONLY SUPERVISION. WE COMPARE ATTENTION-BASED MIL (NO LLM), CLIP-MIL (LLM PROMPTS), AND TWO LLP BASELINES— BAG-LEVEL CROSS-ENTROPY PROPORTION MATCHING (BAGCE) AND DISTRIBUTION MATCHING WITH JENSEN–SHANNON DIVERGENCE (DM-JS). ALL ENCODERS ARE FROZEN FOR FAIRNESS AND ONLY BAG-LEVEL OBJECTIVES ARE OPTIMIZED. NUMBERS ARE MEAN $\pm$ STD OVER 3 RUNS; $\uparrow$ MEANS HIGHER IS BETTER. **NTMP (URE)** ATTAINS THE STRONGEST AP/BEST-F1/AUC, AND THE SIMPLE **ABS** CORRECTION FURTHER STABILIZES AND IMPROVES PERFORMANCE.

Method	AP $\uparrow$	Best-F1 $\uparrow$	AUC $\uparrow$
Attention-MIL (no LLM)	0.85 $\pm$ 0.01	0.80 $\pm$ 0.00	0.91 $\pm$ 0.00
CLIP-MIL (LLM prompts)	0.25 $\pm$ 0.00	0.57 $\pm$ 0.00	0.13 $\pm$ 0.00
LLP–BagCE	0.37 $\pm$ 0.00	0.67 $\pm$ 0.00	0.26 $\pm$ 0.01
LLP–JS	0.38 $\pm$ 0.00	0.67 $\pm$ 0.00	0.27 $\pm$ 0.01
NTMP (URE)	0.92 $\pm$ 0.01	0.84 $\pm$ 0.02	0.86 $\pm$ 0.02
+ ABS correction	0.99$\pm$0.00	0.98$\pm$0.00	0.99$\pm$0.00

g) *Distribution Matching with Jensen–Shannon Divergence (LLP):* Same setup as above but replace the bag-level cross-entropy with the symmetric, bounded JS($\bar{p} \parallel q$), which typically improves robustness under noisy or slightly misspecified bag proportions while keeping supervision at the bag level.

D. Results

Setup and primary comparison.: We first compare test accuracy across methods. **Table IV** summarizes results for $(n, m) = (3, 1)$ under three unlabeled-set priors $\pi \in \{0.2, 0.5, 0.8\}$, with each entry reported as mean $\pm$ std over five independent tuple constructions. Taken together with **Table III**, NTMP(+ABS) not only improves AP/AUROC but also reduces ECE_{TS}/Brier_{TS}, with small Holm-adjusted p -values and positive Cliff’s δ (small-to-medium), indicating statistically significant and practically meaningful gains.

NTMP is best in most settings.: Both corrected variants (ABS/ReLU) consistently lead, often by large margins. For example, on CIFAR-10 with $\pi=0.5$, NTMP(ABS) attains $\approx 80.0\%$ vs. 67.8% for UUcor and 69.2% for plain URE. On MNIST and FASHIONMNIST, NTMP frequently exceeds 95%, whereas UU degrades notably when $\pi \neq \alpha$, highlighting the benefit of exploiting *exact* tuple counts rather than treating $\bar{D}_T$ as a generic unlabeled set.

Plain URE is informative but less reliable.: In a few cases (e.g., SVHN at $\pi=0.2$), URE matches or slightly exceeds corrected NTMP, confirming that the unbiased objective carries signal. However, URE shows higher variance and overfitting,

TABLE VI

UNDER THE SAME EXPERIMENTAL SETTINGS, THE RESULTS PRESENTS THE PERFORMANCE OF DIFFERENT METHODS ACROSS FOUR DATASETS UNDER THE SAME EXPERIMENTAL SETUP, WHERE THE LENGTH OF EACH TUPLE IS SET TO 5 AND EACH TUPLE CONTAINS 2 POSITIVE INSTANCES.

prior	Datasets	P_{NTMPU}	<i>Baseline</i>					
		ABS	NN	URE	UUcor	UU	KM	KM++
0.2	MNIST	96.15±0.17	95.93±0.23	94.28±0.28	94.85±0.27	78.27±1.95	61.09±0.03	61.09±0.01
	FASHIONMNIST	94.68±0.24	91.97±0.24	78.67±0.86	93.14±0.10	94.46±0.47	67.78±0.01	67.89±0.03
	CIFAR10	82.89±1.47	77.43±1.43	78.06±0.64	76.53±0.63	71.41±1.38	51.59±0.00	51.39±0.01
	SVHN	84.35±1.35	65.07±0.76	83.59±1.68	85.25±1.13	78.84±1.20	55.87±0.00	55.80±0.00
0.5	MNIST	97.76±0.41	98.27±0.28	81.13±1.11	61.44±0.60	61.52±1.10	59.21±0.03	59.21±0.01
	FASHIONMNIST	94.08±0.65	96.82±0.14	65.39±0.50	66.53±0.92	61.67±0.53	68.52±0.00	68.54±0.00
	CIFAR10	82.87±1.06	68.16±3.96	77.83±0.87	56.72±1.34	56.09±0.74	51.21±0.00	51.11±0.00
	SVHN	89.12±1.04	82.41±2.05	81.87±1.35	53.49±1.65	51.02±0.00	55.42±0.00	55.45±0.00
0.8	MNIST	95.80±0.17	96.21±0.17	92.89±0.31	92.29±0.35	88.28±0.50	67.40±0.03	67.40±0.00
	FASHIONMNIST	92.40±0.26	91.31±0.34	86.75±0.31	91.78±0.21	83.49±0.17	64.57±0.00	64.69±0.00
	CIFAR10	81.68±2.08	74.73±1.61	79.54±0.59	77.08±0.45	68.32±0.74	52.66±0.00	52.71±0.02
	SVHN	80.83±1.58	83.66±0.67	78.71±1.39	81.90±0.43	75.32±6.95	57.47±0.01	57.68±0.00

TABLE VII

UNDER THE SAME EXPERIMENTAL SETTINGS, WE SET THE LENGTH OF EACH TUPLE TO 7, WITH EACH TUPLE CONTAINING EXACTLY TWO POSITIVE INSTANCE.

prior	Datasets	P_{NTMPU}	<i>Baseline</i>					
		ABS	NN	URE	UUcor	UU	KM	KM++
0.2	MNIST	95.55±0.12	96.23±0.15	95.75±0.15	92.92±0.59	87.86±0.40	63.99±0.00	65.44±0.01
	FASHIONMNIST	93.62±0.21	94.32±0.16	87.31±0.41	92.34±0.30	92.69±0.13	68.73±0.00	68.91±0.00
	CIFAR10	74.82±3.82	77.79±1.94	71.79±2.73	76.34±1.09	76.38±0.70	51.12±0.00	52.64±0.00
	SVHN	72.67±2.18	92.78±0.70	90.15±1.31	85.20±0.95	86.10±2.16	51.33±0.00	51.29±0.01
0.5	MNIST	95.87±0.40	96.96±0.32	84.57±1.55	52.88±0.46	59.20±0.60	64.14±0.01	65.68±0.01
	FASHIONMNIST	87.96±0.76	92.76±0.41	60.10±0.74	51.62±0.45	49.66±1.26	68.36±0.00	68.15±0.01
	CIFAR10	77.90±2.38	71.92±3.29	76.47±2.18	58.42±2.69	57.80±0.94	52.44±0.00	52.62±0.01
	SVHN	81.99±1.15	80.07±2.74	86.96±2.07	53.73±1.29	49.73±0.94	51.59±0.04	51.72±0.02
0.8	MNIST	96.01±0.10	96.23±0.12	93.03±0.37	93.66±0.50	86.72±0.77	63.99±0.00	63.52±0.01
	FASHIONMNIST	93.71±0.16	92.95±0.31	87.20±0.23	91.26±0.07	82.84±0.08	70.88±0.03	71.17±0.01
	CIFAR10	75.70±4.02	77.29±0.77	80.58±0.53	74.80±1.20	69.85±0.65	51.59±0.00	51.78±0.01
	SVHN	80.57±0.91	84.91±1.84	75.27±3.35	83.59±1.56	75.78±6.83	50.46±0.01	50.32±0.04

while ABS/ReLU provide steadier gains across datasets and priors—consistent with our variance and conditioning analysis.

UU(+ABS) > UU, but both trail NTMP: Adding a clamp improves UU over its uncorrected form, yet the gap to NTMP remains substantial—especially around $\pi=0.5$ —suggesting that *count-aware* risk reconstruction outperforms mixing two unlabeled bags that ignore tuple constraints.

Clustering baselines are weakest: KM/KM++ hover around 50–60% in balanced regimes and rarely exceed trivial baselines when imbalanced, confirming that count-only supervision provides signal that unsupervised clustering fails to extract.

Effect of tuple length and clamps.: We repeat the comparison for $n \in \{5, 7\}$ (with appropriate m); see **Tables VI–VII**. Trends persist: NTMP leads consistently, and the advantage grows with larger n (weaker per-instance signal), where exact counts become more valuable. We observe mild, dataset-dependent differences between ABS and ReLU: for $(n, m) = (7, 2)$ ABS sometimes edges out ReLU, whereas for $(3, 1)$ ReLU may slightly win; both effectively reduce

overfitting.

LLM/MIL/LLP baselines (frozen encoders): Table V compares count-only supervision on MNIST tuples with frozen encoders. NTMP (URE) surpasses the MIL and LLP baselines (AP 0.92 ± 0.01 , Best- F_1 0.84 ± 0.02 , AUROC 0.86 ± 0.02). Adding ABS further lifts performance to near ceiling (AP 0.99 ± 0.00 , Best- F_1 0.98 ± 0.00 , AUROC 0.99 ± 0.00). Attention-MIL is competitive (AP 0.85 ± 0.01 , Best- F_1 0.80 ± 0.00 , AUROC 0.91 ± 0.00), whereas the LLM-assisted CLIP-MIL variant lags under these settings. LLP baselines (BagCE/DMJS) provide weaker references. Variances are uniformly small across three runs, indicating a stable ranking under a fixed compute/data protocol.

Prior misspecification sweep (Δ) and stability.: **Figure 4** sweeps $\Delta = |\pi - \alpha|$ and compares NTMP(+ABS) to URE. Panels (a–c): NTMP attains near-saturated Best- F_1 , AUROC, and AP with narrow uncertainty, while URE degrades as Δ increases and exhibits substantially larger variability. Panel (d): the across-seed standard deviation of AP for NTMP remains near zero, while URE’s dispersion is typically 0.2–

0.3. The red dashed line marks Δ_{crit} —the smallest Δ where the 95% CI width of AP exceeds the threshold w^* (cf. Section III-C)—indicating a wide robust window where NTMP remains accurate and stable. Paired Wilcoxon tests on AP confirm NTMP’s superiority at most Δ values (* $p < .05$, ** $p < .01$, *** $p < .001$). These trends persist when varying n at fixed α , suggesting insensitivity to variance induced by tuple length. Overall, the results align with theory: sensitivity scales with the conditioning factor $|\pi - \alpha|^{-1}$, and the ABS/ReLU clamp mitigates finite-sample variance inflation.

Training dynamics and robustness: Figure 3 ($\pi=0.2$, $(n, m)=(3, 1)$) shows that URE achieves high training accuracy quickly and then fluctuates—typical overfitting—whereas clamped objectives stabilize training and improve final test accuracy. Figure 1 reports TPR/FPR/Precision/Macro- F_1 as π varies at fixed $(3, 1)$: TPR and Macro- F_1 are stable or improve with larger π , and FPR decreases markedly, indicating robustness to prior shifts. Figure 2 evaluates three perturbations on MNIST: (a) prior misspecification up to $\pm 30\%$ leaves COR/ F_1 essentially flat and keeps FPR near zero; (b) under synthetic count noise (flip m to $m \pm 1$ with probability Pr), FPR increases approximately linearly in Pr , while COR/TPR/ F_1 decrease monotonically; (c) sweeping π across the near-degenerate regime ($\alpha \approx \pi$) yields small fluctuations except in a narrow band around the boundary. All observations match the theoretical predictions of Sections III-F–III-G. Across the three stressors (prior misspecification, count flips, and $\alpha \approx \pi$), COR/TPR/ F_1 remain stable outside a narrow ill-conditioned band, and FPR grows smoothly under count noise—consistent with the $|\pi - \alpha|^{-1}$ sensitivity predicted by our theory (Fig. 2).

Scope and failure modes vs. LLP/MIL: NTMP is preferable when exact counts (n, m) are available and tuples can be treated as orderless mixtures: the observable pair (tuple aggregate, unlabeled pool) closes a 2×2 system identifiable whenever the effective rate $\bar{\alpha}$ differs from π . LLP requires sufficient cross-bag proportion variation; with nearly identical proportions, the problem becomes ill-posed and high-variance. MIL gains when intra-bag dependencies carry discriminative signal (spatial/temporal structure); it tends to overfit when instances are i.i.d. within a bag and structure is weak, whereas NTMP may underperform if the task fundamentally relies on relations lost under mixture flattening. See Table II for a compact summary.

A setting where structure-aware MIL is expected to win (CIFAR-10/ResNet): Dataset: CIFAR-10. Encoder: ResNet-18 (fine-tuned). Bag construction: for each image, extract $n=8$ spatial crops on a fixed 2×4 grid with 32×32 windows and mild overlaps; define bag label as “at-least-one target” and (optionally) provide a coarse count by the number of crops intersecting the ground-truth object mask (proxy via CAM or a weak saliency detector). Structure: preserve grid indices so neighboring crops are semantically correlated. Models: (i) NTMP with exact count (n, m) but orderless mixing; (ii) LLP with bag proportions; (iii) MIL with gated attention over the 8 crops (optionally a small graph readout with 4-neighbor edges). Protocol: report AP/AUROC/ F_1 and a “shuffle-control” where crop order is randomly permuted and grid indices dropped. **Expectation:** MIL benefits from spatial

coherence and focuses on object-bearing crops; the advantage should diminish under shuffle-control, while NTMP/LLP are relatively insensitive to spatial order—clarifying that MIL’s edge stems from structure rather than supervision volume.

V. CONCLUSION

We studied weakly supervised learning from *tuple-level exact positive counts* (NTMP) and showed how to train a binary classifier without instance labels. Our key contribution is a closed-form *unbiased risk estimator* (URE) that couples the flattened tuple mixture $\tilde{p}_T$ (with weight $\alpha=m/n$) and an auxiliary unlabeled pool p_U (with prior π), eliminating the unknown class-conditionals via a 2×2 linear system. On top of this objective, we established generalization guarantees with optimal $N^{-1/2}$ rates (up to logs), characterized identifiability and conditioning through the factor $|\pi - \alpha|^{-1}$, and derived robustness results for *prior misspecification* and *random count noise* (uniform bias bounds and excess-risk control). We also introduced simple nonnegative *stability corrections* (ABS/ReLU) that reduce variance while preserving asymptotic correctness.

Empirically, NTMP consistently matches or outperforms representative weak-supervision alternatives (e.g., UU variants and clustering) and maintains favorable COR/TPR/FPR/ F_1 trade-offs. The observed trends align with theory: performance remains stable under class-prior shifts and tuple reconfigurations, with degradation confined to the narrow ill-conditioned regime $\alpha \approx \pi$ predicted by the conditioning analysis.

Limitations and future work. (i) Flattening tuples for risk estimation may forgo informative intra-tuple structure; extending NTMP to *tuple-aware* architectures (sets/graphs) under count supervision is a natural direction. (ii) Multi-class and hierarchical-count extensions would broaden applicability. (iii) Jointly learning *prior* and *count* calibrations (e.g., EM or auxiliary calibration losses) could further improve robustness when π and m are uncertain. (iv) Beyond simple clamps, adaptive clipping or control variates may tighten finite-sample variance for URE.

Overall, NTMP shows that coarse *count* supervision, combined with an unlabeled reference and principled risk reconstruction, is a scalable, theoretically grounded, and practically stable alternative where instance-level annotation is costly or infeasible.

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